Adarsh Kumar, PhD

Postdoctoral Fellow, LIMB Lab, Queen's University

Kingston, Ontario, Canada · adarsh.kumar@queensu.ca · Google Scholar · ORCID · babaadarsh.github.io

RESEARCH INTERESTS

Computational and cognitive neuroscience; motor control and sensorimotor learning; state-space, Bayesian, and neural-network modelling of behaviour; multisensory integration; predictive behavioural analytics; neuroplasticity and rehabilitation.

ACADEMIC APPOINTMENTS

Postdoctoral Fellow, LIMB Lab, Queen's University, Canada (2022 to present). Supervisor: Stephen H. Scott.

Research Associate, IIT Gandhinagar, India (2021 to 2022). Supervisor: Pratik Mutha.

EDUCATION

PhD, Mechanical Engineering, IIT Gandhinagar (2015 to 2021). Dissertation: Mechanisms and Neural Substrates Underlying the Acquisition of Effector-Independent Motor Memories (Supervisor: Pratik Mutha).

B.Tech., Biomedical Engineering, NIT Raipur (2011 to 2015).

METHODOLOGICAL EXPERTISE

Computational: computational modelling of learning and behaviour; state-space and Bayesian models; neural-network modelling of behavioural and neural dynamics; predictive modelling; reproducible programming-based analysis and statistics.

Experimental: human sensorimotor adaptation paradigms; KINARM-based motor-control experiments; adaptive psychophysics; multichannel EMG and motor-unit recordings; non-human primate neurophysiology (LFP and multi-electrode arrays); tDCS and TMS protocols.

PEER-REVIEWED PUBLICATIONS

1. Kumar, A.*, Deva Kumar, A.*, Sannamath, S., & Kumar, N. (in press). Contextual cues and transition statistics drive expression of competing motor memories. *iScience*.

2. Kumar, A. D.*, Kumar, A.*, & Kumar, N. (2025). Interaction between model-based and model-free mechanisms in motor learning. *Journal of Neurophysiology*, 134(5), 1714 to 1726.

3. Oza, A.*, Kumar, A.*, Sharma, A., & Mutha, P. K. (2024). Limb-related sensory prediction errors and task-related performance errors facilitate human sensorimotor learning through separate mechanisms. *PLOS Biology*, 22(7), e3002703.

4. Kumar, A., & Mutha, P. K. (2023). Spontaneous recovery in an untrained arm as an assay of interlimb transfer of motor learning. *JEP: Human Perception and Performance*, 49(5), 725.

5. Sadaphal, D., Kumar, A., & Mutha, P. K. (2022). Sensorimotor learning in response to errors in task performance. *eNeuro*, 9(2).

6. Kumar, A., Panthi, G., Divakar, R., & Mutha, P. K. (2020). Mechanistic determinants of effector-independent motor memory encoding. *PNAS*, 117(29), 17338 to 17347.

7. Kumar, N., Kumar, A., Sonane, B., & Mutha, P. K. (2018). Interference between competing motor memories developed through learning with different limbs. *Journal of Neurophysiology*, 120(3), 1061 to 1073.

* equal contribution

PREPRINTS (UNDER REVIEW)

1. Pathak, A., Kumar, A., Welsh, T., & Mutha, P. K. (2026). Interference between motor memories arises from implicit recalibration. *bioRxiv*.

2. Kumar, A. D., Kumar, A., & Kumar, N. (2026). Consolidation separates implicit and explicit components of compound motor memories. *bioRxiv*.

3. Suresh, T., Kumar, A., & Mutha, P. K. (2025). Reactivation protects human motor memories against interference from competing inputs. *bioRxiv*.

AWARDS & FELLOWSHIPS

Diversity Fellowship, Neural Control of Movement, Kobe, Japan (2026). Postdoctoral Conference Grant, Queen's University (2024).

TEACHING

Guest lecture, Introduction to Motor Learning, Faculty of Kinesiology and Physical Education, University of Toronto (KPE160H). Teaching assistantships at IIT Gandhinagar (2015 to 2019): Advanced Probability Theory, Control Systems, Modern Control Theory, Fundamental Neuroscience, Industrial Engineering and Operations Research, and others.

PROFESSIONAL SERVICE

Reviewer: Journal of Neurophysiology; Journal of Neuroscience; Cerebral Cortex; PLOS Biology; eNeuro; Motor Control; npj Microgravity. **Leadership:** PhD Senator, IIT Gandhinagar (2018 to 2020); President, Student Association, NIT Raipur (2013 to 2015).

OPEN-SOURCE SOFTWARE

Six tested toolboxes spanning behaviour, modelling, EMG, kinematics, neural decoding, and recurrent-network dynamics. github.com/BabaAdarsh

SELECTED COLLABORATORS

Pratik Mutha (IIT Gandhinagar) · Stephen H. Scott (Queen's) · Neeraj Kumar (IIT Hyderabad) · Tim Welsh (Toronto) · Numa Dancause (Montreal) · D. J. Cook (Queen's).